ZEKUN LI

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EDUCATION

Brown University **Providence, RI, USA**
Ph.D. Student, Computer Science August 2023 - May 2027 (expected)
Supervisor: Prof. Srinath Sridhar
Research Area: Human-centric Vision Content Generation, Unified Multimodal Learning

University of Electronic Science and Technology of China **Chengdu, Sichuan, China**
Bachelor of Engineering with honor, Computer Science and Technology September 2019 - July 2023
GPA: 3.78/4.0
UESTC Outstanding Undergraduate Thesis Awards

PREPRINT & IN-SUBMISSION

[1] **Ms. Forcing: Efficient Streaming Video Generation with Multi-Scale Patchification and Attention**
Zekun Li*, Xiaoyan Cong*, Hongyu Li, Zhiyang Dou, Chuan Guo, Abhay Mittal, Sizhe An, Srinath Sridhar

[2] **World-T^3: Test-Time Training as Persistent Memory for Real-Time Video World Models**
Junyi Wu*, Zekun Li*, Chenhongyi Yang, Chen Guo, Lingchen Yang, Yongan Zhang, Xinyi Zhang, Hongbing Hu, Bartek Rymkowski, Srinath Sridhar, Yan Yan, Xijie Huang

[3] **HumanCLAW: Can Vision-Language Models Act Through a Body?**
Siyao Li*, Jiawei Gu*, Shuai Liu, Kairui Hu, Zekun Li, Chengcheng Tang, Po-Chen Wu, Ivan Shugurov, Lingjie Li, Lingni Ma, Lingling Tao, Michael Zollhoefer, Sizhe An, Abhay Mittal, Amy Zhao, Ranjay Krishna, Manling Li, Ziwei Liu, Chuan Guo

[4] **MoSAT: Human Motion Generation from Spatial Audio and Textual Description**
Shuyang Xu, Zhiyang Dou, Yiduo Hao, Zekun Li, Liang Pan, Jingbo Wang, Cheng Lin, Yuan Liu, Wenping Wang, Mingmin Zhao, Taku Komura

[5] **Sonic-AMG: Streaming Human Motion Generation from Spatial Audio**
Jingcheng Ni, Zekun Li, Zhiyang Dou, Xiaoyan Cong, Ziyao Zeng, Hongyang Du, Taku Komura, Srinath Sridhar

[6] **EgoReAct: Egocentric Video-Driven 3D Human Reaction Generation**
Libo Zhang, Zekun Li, Tianyu Li, Zeyu Cao, Rui Xu, Xiao-Xiao Long, Wenjia Wang, Jingbo Wang, Yuan Liu, Wenping Wang, Daquan Zhou, Taku Komura, Zhiyang Dou

[7] **IAM: Identity-Aware Human Motion and Shape Joint Generation**
Wenqi Jia, Zekun Li, Abhay Mittal, Chengcheng Tang, Chuan Guo, Lezi Wang, James Matthew Rehg, Lingling Tao, Sizhe An

PUBLICATION

[8] **UMO: Unified In-Context Learning Unlocks Motion Foundation Model Priors** [ECCV2026]
Xiaoyan Cong*, Zekun Li*, Zhiyang Dou, Hongyu Li, Omid Taheri, Chuan Guo, Abhay Mittal, Sizhe An, Taku Komura, Wojciech Matusik, Michael J. Black, Srinath Sridhar

[9] **Deform360: A Massive Multi-view Visuotactile Dataset for Deformable World Models** [ECCV2026]
Hongyu Li, Wanjia Fu, Xiaoyan Cong, Zekun Li, Binghao Huang, Hanxiao Jiang, Xintong He, Yiqing Liang, Rao Fu, Tao Lu, Srinath Sridhar, Kevin A. Smith, George Konidaris, Yunzhu Li

[10] **LLaMo: Scaling Pretrained Language Models for Unified Motion Understanding and Generation with Continuous Autoregressive Tokens** [CVPR2026]
Zekun Li, Sizhe An, Chengcheng Tang, chuan guo, Ivan Shugurov, Linguang Zhang, Amy Zhao, Srinath Sridhar,

Lingling Tao, Abhay Mittal

[11] **PackUV: Packed Gaussian UV Maps for 4D Volumetric Video** [CVPR2026]
Aashish Rai, Angela Xing, Anushka agarwal, Xiaoyan Cong, Zekun Li, Tao Lu, Aayush Prakash, Srinath Sridhar

[12] **VideoGPA: Distilling Geometry Priors for 3D-Consistent Video Generation** [ICML2026]
Hongyang Du*, Junjie Ye*, Xiaoyan Cong*, Runhao Li, Jingcheng Ni, Aman Agarwal, Zeqi Zhou, Zekun Li, Randall Balestriero, Yue Wang

[13] **GenHSI: Controllable Generation of Human-Scene Interaction Videos** [WACV2026]
Zekun Li, Rui Zhou, Rahul Sajnani, Xiaoyan Cong, Daniel Ritchie, Srinath Sridhar

[14] **Surf-D: Generating High-Quality Surfaces of Arbitrary Topologies Using Diffusion Models** [ECCV2024]
Zhengming Yu, Zhiyang Dou, Xiaoxiao Long, Cheng Lin, Zekun Li, Yuan Liu, Norman Müller, Taku Komura, Marc Habermann, Christian Theobalt, Xin Li, Wenping Wang

[15] **MANUS: Markerless Grasp Capture using Articulated 3D Gaussians** [CVPR2024]
Chandradeep Pokhariya, Ishaan Nikhil Shah, Angela Xing, Zekun Li, Kefan Chen, Avinash Sharma, Srinath Sridhar

[16] **Learning Anchor Transformations for 3D Garment Animation** [CVPR2023]
Fang Zhao, Zekun Li, Shaoli Huang, Junwu Weng, Tianfei Zhou, Guosen Xie, Jue Wang, Ying Shan

[17] **Eliminating Gradient Conflict in Reference-based Line-Art Colorization** [ECCV2022]
Zekun Li, Zhengyang Geng, Zhao Kang, Wenyu Chen, and Yibo Yang

EXPERIENCE

Meta Reality Lab *Research Intern*
May 2026 - August 2026 Supervisor: Dr. Alexander Richard
♢ Project: 3D Social Avatar Behavior Generation with Streaming OmniDiT (Audio, Face, and Body)
- Curated 60K+ hours of heterogeneous, partially paired audio, face, upper body gesture, full body locomotion data including real-world capturing and data synthesis.
- Developed full-stack curriculum-based bidirectional pretraining and causal distillation to learn from incomplete multimodal large scale dataset for real-time streaming.

Meta Reality Lab *Research Intern*
May 2025 - August 2025 Supervisor: Abhay Mittal
♢ Project: Unified Generation and Understanding of 3D Human Motion with Continuous-Token Autoregressive
- Proposed a unified motion-language large model trained from an LLM while preserving language performance.
- Adopting continuous token autoregressive to achieve high-fidelity motion generation with real-time streaming.

Honda Research Institute *Research Intern*
June 2024 - August 2024 Supervisor: Enna Sachdeva
♢ Project: Uncertainty-aware Human-Object Interaction Tracking from Monocular Video
- Proposed an autoregressive tracking method for the poses of the human and object with uncertainty.
- Leveraging the visibility and physics constraints for post-sampling to eliminate the accumulated error in human-object interaction tracking.

AI Lab, Tencent *Research Intern*
October 2022 - June 2023 Supervisor: Prof. Fang Zhao
♢ Project: Learning-based 3D Garment Animation
- Proposed AnchorDEF to predict 3D garment animation based on body motion sequence inputs.
- Jointly learned topology-aware anchors and residual deformations, achieving SOTA results, especially for loose-fitting garments.

Cognitive Computing and Intelligent Decision Lab, UESTC *Research Assistant*
September 2020 - September 2022 Supervisor: Prof. Zhao Kang
♢ Project: Reference-based line-art colorization

- Identified gradient conflicts in SDPA and proposed Stop-Gradient Attention to stabilize cross-modal learning.
- Reduced FID by up to 27.21% and improved SSIM by up to 25.67% across multiple benchmarks, outperforming prior reference-based colorization modules.

PROFESSIONAL SERVICE

Conference Reviewer: CVPR 2024, SIGGRAPH 2024, ECCV2024, AAAI2025, ICLR2025, ECCV2026,SIGGRAPH ASIA 2026, WACV 2027
Google explore CSR: Ph.D. mentor 2024